

Performance Testing

Date	18 July 2026
Team ID	TBD
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Performance Testing, Response Time Testing
Target Module / API	Speech Evaluation Module, Whisper Transcription, Gemini AI Feedback API
Test Environment	Local System (Windows 11, Python 3.11, Streamlit)
Test Date	18 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Audio Upload Testing	1	20 sec	Audio uploaded successfully
2	Speech Evaluation	1	60 sec	Transcript and similarity generated
3	Gemini AI Feedback	1	45 sec	AI feedback generated successfully

4	Dashboard Loading	1	15 sec	Dashboard loads without delay
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 sec	Pass	Within limit
2	Response Time (Max)	< 5 seconds	3.8 sec	Pass	Acceptable
3	Throughput (Req/sec)	10 Req/sec	12 Req/sec	Pass	Good
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	52%	Pass	Normal
6	Memory Utilization	< 80%	60%	Pass	Stable

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

